



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Adaptive Login Fraud Detection and Risk Scoring System	
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Abstract:

Fraudulent login attempts are a major cybersecurity concern in modern web applications, where attackers may gain unauthorized access by using stolen credentials, unfamiliar devices, unusual locations, or abnormal login behavior. Conventional authentication mechanisms primarily depend on usernames, passwords, and fixed security rules, which may not be sufficient to identify sophisticated or continuously evolving login threats. Therefore, there is a need for an intelligent and adaptive system that can analyze login behavior and dynamically determine the risk associated with each authentication attempt.

The proposed AI-Driven Adaptive Login Fraud Detection and Risk Scoring System aims to detect suspicious login activities by analyzing behavioral and contextual characteristics associated with user authentication. The system collects relevant login attributes such as login time, device information, IP address, location, failed login attempts, and historical login patterns. These attributes are processed to identify deviations from normal user behavior. A machine learning-based anomaly detection approach is incorporated to classify login attempts and generate a fraud risk score ranging from low to critical risk.

The proposed system follows a Service-Oriented Architecture using independent microservices to provide modularity, scalability, maintainability, and secure communication between components. The major services include an Authentication Service, Fraud and Risk Analysis Service, Alert and Notification Service, and Security Analytics Service. The Authentication Service manages user registration, secure login, JWT-based authentication, and role-based authorization. The Fraud and Risk Analysis Service evaluates login behavior and communicates with the machine learning model to calculate the risk score. Based on the calculated risk level, the system can allow a low-risk login, request additional verification for a medium-risk login, or block and report a high-risk login. The Alert and Notification Service manages suspicious-login alerts, while the Security Analytics Service provides administrators with a centralized dashboard for monitoring authentication activities and security events.

The system is implemented using modern microservices technologies including Spring Boot, Spring Cloud Gateway, Eureka Service Registry, JWT, REST APIs, database-per-service architecture, and machine learning components. API testing, unit testing, integration testing, security analysis, and CI/CD practices are incorporated to improve the reliability and security of the application. The proposed approach aims to provide a more intelligent alternative to static authentication mechanisms by enabling adaptive security decisions based on the real-time risk associated with each login attempt.

Project Architecture/Project Overview:



